Risk Analysis in Psychological Emergencies through Latent Variable Models

Application of Rasch Model and Latent Class Analysis

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Agenda

1 Introduction and Context

2 Theoretical Framework

3 Methodology

4 Results

5 Discussion and Implications

6 Conclusions

Project Context

Research Project

- **Director**: Lic. Carolina Rojo
- **Focus**: Emergencies of the 911 Psychology Service
- **Institution**: UCASAL

Problem

- Need to evaluate psychological emergencies
- Lack of standardized instruments
- Optimization of 911 response

Proposed Solution

Development of the **General Protocol for Psychological Emergencies**

- Standardized evaluation
- Solid psychometric foundation
- Practical application in emergencies

Challenge

Validate the protocol through advanced statistical models

Study Objectives

1. **Evaluate psychometric properties**
   - Apply latent variable models
   - Analyze item fit
2. **Explore dimensionality**
   - Rasch model
   - Mixed Rasch model
3. **Identify latent structures**
   - Latent class analysis
   - Risk profiles
4. **Optimize the protocol**
   - Reduce redundant items
   - Improve efficiency

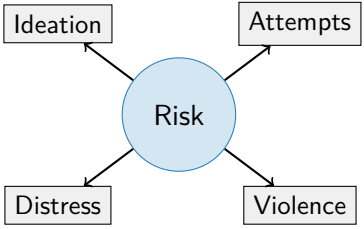
Latent Variables in Psychology

Definition

Constructs not directly observable that are inferred from manifest variables

Examples

- Intelligence
- Personality
- Attitudes
- **Psychological risk**



The Rasch Model

Foundations

- Unidimensional latent trait model
- Response probability depends on:
  - Person level (θ)
  - Item difficulty (β)

Model Equation

$$P_{is} = \frac{e^{(\theta_s - \beta_i)}}{1 + e^{(\theta_s - \beta_i)}} \tag{1}$$

Assumptions

- **Unidimensionality**: A single underlying trait
- **Local independence**: Independent responses given the trait

Mixed Rasch Model

When to use it?

- When the simple model doesn't fit
- Heterogeneous population
- Multiple latent subgroups

Advantages

- Identifies homogeneous classes
- Valid Rasch model within each class
- Class-specific parameters

In this study

Allows identification of different types of psychological emergencies with distinct response patterns

Study Design

Characteristics

- Cross-sectional observational study
- Psychometric analysis
- Quantitative approach

Sample

- **N = 310** protocols
- Real calls to 911
- Psychology Service
- Salta Province

Distribution by Classes

Class	Percentage
Class 1	74%
Class 2	20%
Class 3	6%

Note

Real data from population contacting the emergency service

Instrument: 12-Item Protocol

Suicide Risk

- Current attempt
- Previous attempts
- Suicidal ideation
- Self-harm

Context

- Addiction
- Abandonment
- Separation
- Boundary difficulties

Current State

- Psychiatric patient
- Sexual problems
- Distress
- Violence episodes

Format

All items are dichotomous (Yes/No)

Analysis Procedure

Stage 1: Latent Class Analysis

- Identify optimal number of classes
- BIC criterion

Stage 2: Rasch Model

- Evaluate global fit
- Q-index and RMSEA

Stage 3: Mixed Rasch Model

- Apply to identified classes
- Evaluate unidimensionality by class

Stage 4: Optimization

- Remove problematic items
- Re-evaluate model

Latent Class Analysis

Number of Classes	CAIC	BIC
1	3003	2991
2	2911	2886
3	2923	**2885**
4	2959	2908

Decision

The **3-class** model is selected for presenting the lowest BIC

Characterization of the Three Classes

Characteristic	**Class 1 (74%)**	**Class 2 (20%)**	**Class 3 (6%)**
Profile	Low risk	Suicide risk	High risk
Distress	0.42	0.35	**0.94**
Violence episodes	0.35	0.12	**0.81**
Suicidal ideation	0.08	**0.55**	**0.75**
Current attempt	0.02	**0.50**	0.44
Psychiatric patient	0.28	0.15	**0.63**

Interpretation

- **Class 1**: Lower severity emergencies
- **Class 2**: Focus on suicide risk
- **Class 3**: Multiple risk indicators

Rasch Model - Initial Fit

Items with Poor Fit

- Current suicide attempt (Q=0.402)
- Abandonment (Q=0.511)
- Boundary difficulties (Q=0.438)
- Psychiatric patient (Q=0.425)
- Sexual problems (Q=0.585)

Problem

- RMSEA = 0.383
- No unidimensionality
- 5 of 12 problematic items

Strategy

Remove items with Q-index > 0.30 and apply mixed Rasch model

Mixed Rasch Model - Results

Optimized model with 6 items

Conceptually relevant items were maintained

Class	n (%)	RMSEA	Fit
1	254 (82%)	0.106	Acceptable
2	43 (14%)	0.174	Mediocre
3	13 (4%)	2.268	Rejectable

Note

Class 3 presents problems due to reduced sample size (n=13)

Two Identified Dimensions

Dimension 1

Violence toward the environment

- Distress
- Violence episodes
- **82% of the sample**

Dimension 2

Violence toward oneself

- Suicidal ideation
- Current attempt
- Previous attempts
- **18% of the sample**

Convergence with Factor Analysis

Results coincide with a previous exploratory factor analysis that identified similar factors

Theoretical Implications

Main Findings

1. **Multidimensionality**
   - Psychological emergencies are NOT unidimensional
   - At least two differentiated types exist

2. **Population heterogeneity**
   - Different profiles of 911 users
   - Distinct intervention needs

3. **Construct validity**
   - Convergence between statistical methods
   - Empirical support for dimensions

Practical Implications

Protocol Optimization

- From 12 to 6 effective items
- Greater efficiency
- No information loss

Training

- Type-specific training
- Differentiated protocols
- Better response

Type 1 Protocol

Violence toward oneself

- Suicide risk evaluation
- Immediate referral
- Follow-up

Type 2 Protocol

Violence toward environment

- Initial containment
- Crisis management
- Context evaluation

Study Limitations

1. **Sample size**
   - Class 3 with only 13 cases
   - Limits generalizations
2. **Dichotomous nature**
   - Loss of information about severity
   - Simplification of symptoms
3. **Specific context**
   - Salta 911 system
   - Regional characteristics
4. **No external validation**
   - Lack of case follow-up
   - No external criteria

Future Directions

Research

- Develop polytomous versions of the protocol
- Longitudinal follow-up studies
- Validation in other contexts
- Integration with crisis theories

Practice

- Implementation of optimized protocol
- Automatic classification system
- Specific intervention protocols
- Effectiveness evaluation

Main Conclusions

1. 911 psychological emergencies present a **bidimensional structure**:
   - Violence toward oneself
   - Violence toward the environment
2. The protocol can be **optimized** from 12 to 6 items while maintaining its validity
3. There is **convergence** between the mixed Rasch model and factor analysis
4. The majority of emergencies (82%) correspond to a **lower severity profile**
5. 14% present **high suicide risk** requiring specialized intervention

Final Message

Main Contribution

The application of advanced psychometric models allows a deeper understanding of psychological emergencies, facilitating the development of more effective and efficient interventions

Expected Impact

Improvement in the quality of care of the 911 Psychology Service through evidence-based protocols

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Thank you!

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Questions?